

Automatic Reflection Level Classification in Hungarian Student Essays

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Abstract

Reflective thinking is a key competency in education, but assessing reflective writing remains a time-consuming and subjective task for education experts. While automated reflective analysis has been explored in several languages, Hungarian language was not researched extensively. In this paper, we present the first comprehensive study on automatic reflection level classification in Hungarian student essays. We used a large, expert-annotated Hungarian dataset consisting of 1,954 reflective essays collected over multiple academic years and labeled on a four-level reflection scale. We investigate two approaches: (1) classical machine learning models using TF-IDF and semantic embedding features, and (2) Hungarian-specific transformer models fine-tuned for document-level reflection classification. To address the strong class imbalance in the dataset, we systematically examine class weighting, oversampling, data augmentation, and alternative loss functions. An extensive ablation study is conducted to analyze the contribution of each modeling and balancing strategy. Our results show that shallow machine learning models with appropriate feature engineering achieve strong overall performance, reaching up to **71%** overall score averaged

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over accuracy, F1-score, and ROC AUC metrics, while transformer-based models achieve slightly lower overall score (**68%**) averaged over the same metrics, but demonstrate better generalization on minority reflection classes. These findings highlight the continued relevance of classical methods for low-resource settings and the robustness of transformer models for imbalanced classification. The proposed dataset and experimental insights provide a solid foundation for future research on automated reflective analysis in Hungarian and other morphologically rich languages.

Keywords: Reflective analysis; Imbalanced learning; Natural language processing; Hungarian text classification

1 Introduction

Critically analyzing situations and trying to understand them better is a key skill in human cognition. Reflective thinking is widely recognized as a core educational competency in international frameworks and policy documents (Tuning Academy, 2024; OECD, 2019; Official Journal of the European Union, 2018). In higher education and professional training, reflective writing is commonly used to encourage deeper learning and professional development (Lee, 2005; Korthagen and Nuijten, 2022; Lim et al., 2023; Newton, 2000; Occhiuto et al., 2024). However, evaluating reflective texts is typically a manual process and therefore difficult to scale in educational practice (Dyment and O’Connell, 2011; Rogers et al., 2019; Ullmann, 2019).

To address this limitation, recent research has explored automatic reflective text analysis using both classical machine learning and transformer-based natural language processing approaches (Ullmann, 2019; Alrashidi et al., 2022; Alrashidi and Ullmann, 2024; Wulff et al., 2023; Zhang et al., 2024; Nehyba and Štefánik, 2023; Chong et al., 2020). These studies suggest that automated reflection classification is feasible across multiple settings and annotation schemes. At the same time, existing work is concentrated mainly on English, German, or multilingual datasets, while Hungarian remains underexplored for document-level reflection classification (Wulff et al., 2023; Zhang et al., 2024; Nehyba and Štefánik, 2023). This gap is important because Hungarian is a morphologically rich language, and Hungarian NLP benefits from language-specific resources and model development (Nemeskey, 2021; Yang et al., 2023).

This study investigates automatic reflection level classification in Hungarian student essays using two model families: (1) classical machine learn-

ing with TF-IDF and semantic embeddings, and (2) Hungarian transformer models fine-tuned for document classification. Because our dataset is strongly imbalanced, balancing strategies were evaluated systematically, and an ablation study was conducted to quantify the contribution of each modeling choice.

Our contributions are the following:

1. A comprehensive Hungarian analysis of automatic reflection-level classification in student essays, using a large expert-annotated corpus collected across multiple academic years.
2. Comparing two modeling paradigms for Hungarian reflective text classification: classical machine learning with engineered features and fine-tuned Hungarian transformer models.
3. Systematic evaluation of multiple imbalance-handling techniques to improve minority-class performance.
4. An ablation study that clarifies the effect of model family and balancing strategy on classification performance.

The rest of the paper will introduce the related works in Section 2, followed by the methodology in Section 3. Experimental results are presented in Section 4, followed by a discussion in Section 5, and conclusions in Section 6.

2 Related Work

This section reviews relevant literature in two main areas. Firstly, it examines the previous literature about the importance of reflective thinking in the field of education. Secondly, it explores the current state of the automated reflective text classification systems, highlighting key methodologies and findings from prior research.

2.1 Reflective thinking in education

Multiple international frameworks such as the Tuning project (Tuning Academy, 2024), the OECD Learning Compass 2030 (OECD, 2019) and the EU Key Competences (Official Journal of the European Union, 2018) identify reflective thinking or critical reflection as a generic, transformative or

transversal competency, respectively, and require educational programmes to incorporate it into learning outcomes.

Reflective thinking is the deliberate process of analyzing and evaluating one’s own thoughts, experiences, and actions to gain deeper insights and improve future decision-making. In university settings, reflective thinking is often explored in the context of learning, for example, by reviewing one’s own understanding of the material presented in theoretical courses and critically comparing it with personal beliefs or practical experiences. Reflection also enables students to interpret their experiences during practical placements, fostering deeper learning and professional development. In many professional fields, reflection is employed during the practical training phase to support the development of students’ professional skills and the formation of their professional identity. This is evident in education, in teacher training (Lee, 2005; Káplár-Kodácsy and Dorner, 2020; Korthagen and Nuijten, 2022), or in early childhood educator programs (Sumsion, 2000; Cherrington, 2018), as well as in medical education (Lim et al., 2023; Wald and Reis, 2010; Wald et al., 2019), nursing programs (Tashiro et al., 2013; Newton, 2000), and social work training (Occhiuto et al., 2024).

In professional training settings, as well as in the context of lifelong learning, reflection is viewed as a means of continual personal and professional development because it enables the reflective thinker to withstand habitual action and to seek solutions to problems which may be weakly defined or ambiguous (Ullmann (2019)) and arise during human encounters.

Reflective writing is a pedagogical practice that can foster reflective thinking, for example through journaling. Its summarizing end product, the reflective essay, can serve as a record of the reflective thinking process and can therefore be formally assessed by teachers, using for instance rubrics.

2.2 Automated reflective text classification systems

Automated text classification systems have been widely studied in various domains, including sentiment analysis (Pilicita-Garrido and Barra, 2025; Grimalt-Álvaro and Usart, 2024), topic categorization (Schulman and Barbosa, 2018; Tang and Cao, 2015), and educational text analysis (Ferreira-Mello et al., 2019). In the context of reflective text classification, several studies have explored the use of machine learning techniques to classify reflective essays based on different aspects.

Ullmann (2019) conducted a foundational study on the automated analysis of reflective writing using machine learning approaches. The research

evaluated 76 student essays mainly from second- and third-year health, business, and engineering programs, annotated through a large-scale crowdsourcing process involving thousands of contributors. The study adopted a comprehensive reflection model using eight categories: one depth dimension (reflection vs. non-reflection) and seven breadth dimensions, including description of experience, feelings, personal belief, awareness of difficulties, perspective, lessons learned, and future intentions. Using unigram features, several supervised learning algorithms were tested, including Support Vector Machines, Random Forest, Naïve Bayes, and Neural Networks. The models achieved accuracies between 70% and 90% (Cohen’s $K = 0.53-0.85$), approximately 10% lower than manual annotation performance. This work demonstrated that reflective categories in student essays can be reliably identified through automated means, providing a scalable foundation for reflection analytics in educational contexts.

As a continuation of the previous publication, Alrashidi et al. (2022) proposed an automated reflective writing analysis system to classify reflection indicators in computer science students’ essays using machine learning and natural language processing. The study applied the Reflective Writing Framework (RWF), focusing on the seven breadth categories of reflection as in the previous publication excluding the depth dimension of reflection. Using 74 student essays comprising 1,113 annotated sentences, the authors extracted linguistic features based on n-grams and part-of-speech (PoS) n-grams and employed a Random Forest classifier for binary detection of each indicator. The approach achieved accuracies ranging from 75% to 96% (Cohen’s $K = 0.17-0.67$), demonstrating that such linguistic representations can effectively capture reflection-related features and enable automated assessment of reflective depth in student writing.

Alrashidi and Ullmann (2024) investigated the automatic classification of reflection depth in STEM student writings, focusing on Computer Science (CS) education. Building on the seven reflection indicators defined by Ullmann (2019) - description of experience, understanding, feelings, reasoning, perspective, new learning, and future action - the study explored both the breadth and depth dimensions of reflection. Using a dataset of 1,200 annotated sentences from CS students, the authors extracted four types of linguistic features: n-grams, part-of-speech (POS) n-grams, sentiment features (from AFINN, Bing, and NRC lexicons), and WordNet-Affect features. They experimented with a wide range of machine learning algorithms, including Random Forest, Support Vector Machine (SVM), Naïve Bayes, Bayesian Network, Logistic Regression, Neural Network, and Deep Learning models. A comparative evaluation also included the transformer

model XLM-RoBERTa (Conneau et al., 2020) for cross-lingual reflection detection. Their proposed two-stage framework-first classifying reflection indicators, then predicting the depth of reflection (non-reflective, reflective, critically reflective) - achieved higher performance (Cohen’s $K = 0.70$, accuracy = 82.85%) than single-stage feature-based approaches. This work demonstrated that using breadth indicators as features significantly improves the automated detection of reflection depth and represents one of the first attempts to automate reflective writing assessment within CS education using machine learning and natural language processing (NLP).

Solopova et al. (2023) presents PapagAI, one of the first open-source systems for automated feedback on reflective essays, implemented as a hybrid AI approach combining symbolic and neural methods. The system processes student reflections (drawn from a German reflective corpus) and extracts a rich set of signals: it classifies emotional content at the sentence level using a fine-tuned RoBERTa (Liu et al., 2019); it recognizes which Gibbs reflective cycle phase (description, feelings, evaluation, analysis, conclusion, or action plan) a given sentence belongs to via an ELECTRA (Clark et al., 2020) multi-class classifier; it assesses the overall “reflective level” of the text (e.g. descriptive, dialogical, transformative) using a BERT large model; it also applies BERTopic (Grootendorst, 2022) on sentence embeddings for topic modeling; and it derives linguistic features (e.g. sentence length, subordinate clause counts, discourse markers) via spaCy and RFTagger (Schmid and Laws, 2008). These multiple signals are combined via a rule-based reasoner that selects from predefined feedback templates (prompts) to produce tailored formative comments. The authors contrast PapagAI with generative LLM approaches (such as GPT-style models), arguing that the hybrid design offers advantages in transparency, controllability, and alignment with didactic theory, while acknowledging tradeoffs in flexibility and expressiveness.

Zhang et al. (2024) present a comprehensive comparison of shallow machine-learning models and pre-trained transformer models for document-level classification of reflective writing, using a dataset of 1,043 reflections collected from a German teacher-education program. The authors evaluate sparse/textual representations (bag-of-words and term frequency-inverse document frequency as TF-IDF) and psycholinguistic features derived from Pennebaker et al. (2015), feeding these features into several classical classifiers (e.g., Ridge, XGBoost, SGD). They contrast these baselines with fine-tuned transformer families BERT, RoBERTa and long-context transformers like Longformer (Beltagy et al., 2020) and BigBird (Zaheer et al., 2021). Their results show a clear advantage for models that can process

extended contexts: Longformer and BigBird achieved the highest classification accuracies ($\approx 77\%$), while shallow models remained below $\approx 60\%$ accuracy. The study therefore highlights the importance of document-level modeling and long-context capacity when automatically assessing reflective texts, and it underscores how modern transformer architectures improve the fidelity of automated reflection classification compared to traditional sparse or psycholinguistic feature approaches.

Wulff et al. (2023) investigate the use of segment-level classification to automatically detect reflective categories in preservice physics teachers’ writings, using a fine-tuned BERT model. Their corpus has 270 reflections from 92 preservice teachers, each divided into segments that express a single cohesive idea. Drawing on reflection support models e.g. (Korthagen and Vasalos, 2005; Ullmann, 2019), the authors label units into five canonical reflective elements: circumstances, description, evaluation, alternative devising, and derivation of consequences. In addition to BERT, they implement feedforward neural networks and LSTM baselines, and compare with ELMo (Peters et al., 2018) + SVM and simpler count-based representations. Their models are carefully tuned via grid search and evaluated under sensitivity analyses (sampling fractions of training data). They apply Integrated Gradients via Captum (PyTorch team, 2025) to highlight token contributions in classification. The results indicate that BERT significantly outperforms the baselines, and that interpretability methods can enhance the pedagogical transparency of automated reflection classification.

Nehyba and Štefánik (2023) present Applications of Deep Language Models for Reflective Writings, a robust study of reflective writing analysis in teacher education contexts, employing deep pretrained models in a multilingual setting. They detail a strict annotation process of student-teachers’ reflective journal sentences, and experiment with both traditional machine-learning representations (e.g., bag-of-words) and deep models (notably XLM-Roberta). Their results indicate strong performance (accuracy from $\approx 76.6\%$ up to $\approx 100\%$ in high-confidence subsets) and suggest the approach is applicable across more than 100 languages with minimal loss in accuracy. The study underscores the practical value of pretrained architectures for capturing reflective thinking in texts, emphasises annotation and multilingual scalability, and provides an open-source resource framework. The present work builds on this by exploring a Hungarian-language domain and investigating whether a language-specific pretrained model can outperform or complement a multilingual model in reflective-writing classification.

Chong et al. (2020) explore the use of natural-language-processing (NLP) and fuzzy logic for analysing students’ reflective writing in a psychology

course. Their corpus comprises 47 reflective journals, collected via end-of-semester surveys, and pre-processed using NLTK (NLTK team, 2025) for cleaning and tokenisation. The authors build a domain-specific keyword database and apply a fuzzy logic system that uses count-based features (keyword counts, sentiment-oriented terms) to categorise each journal at a document level into predefined reflection-levels. Their analysis finds that the majority of student writings fall into lower levels of reflection (levels 1-2). Although the methodology is relatively lightweight compared to more recent deep-learning-based approaches, this work demonstrates the value of combining domain lexica and rule-based logic in automated reflection analytics. It thus provides an early proof-of-concept in this space, and serves as a useful baseline when comparing more sophisticated ML/NLP pipelines for reflective writing classification.

3 Methodology

3.1 Dataset

The dataset used in this study was collected by experts in the field of education and reflective writing at a large public University in Central Europe. They collected essays in Hungarian from students who completed their studies over a period of 4 consecutive years. Throughout the six semesters of the Early Childhood Education programme, students complete pedagogical practice placements in nurseries, where they receive field-based training. At the end of each semester, they are required to write a reflective essay addressing guiding questions related to their practical experiences. To support the preparation of these essays, students receive detailed written guidance (based on the work of Gyöngy (2021)), including an assessment rubric that can be used for self-evaluation prior to submitting the final work.

Experts rated the essays on multiple dimensions (formal, structural, and content-related), including reflection level, format, and text creation. Only reflection-level labels are used in this study. The dataset contains almost 1954 annotated essays from roughly 450 students. The dataset is not publicly available due to privacy and ethical considerations.

The distribution of the reflection levels in the dataset is heavily imbalanced, as demonstrated in Figure 1. The reflection levels are rated on a scale from 0 to 3 where 0 means no reflection, 1 means low reflection, 2 means medium reflection and 3 means high reflection.

Ethical permission for this research was granted by the ethics committee of Anonymous University(Number).

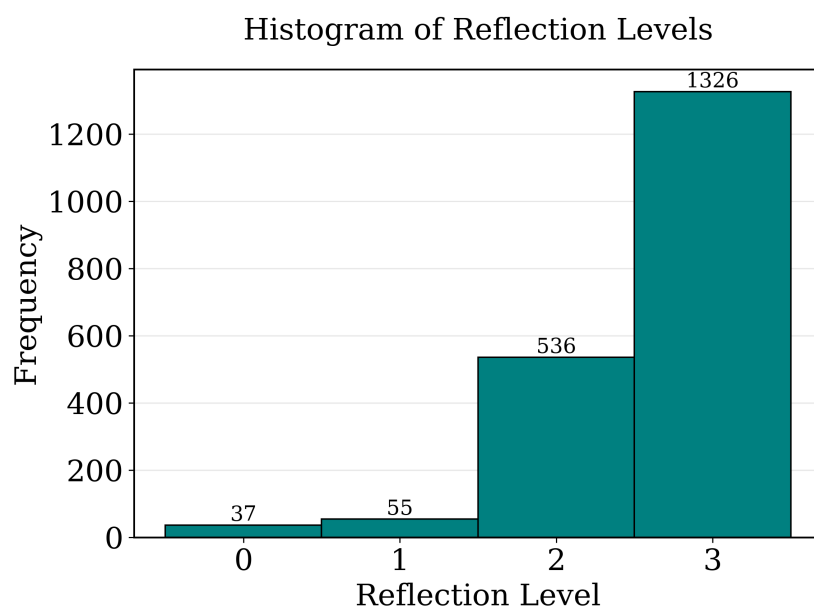


Figure 1: Distribution of reflection levels in the dataset. From 0 (no reflection) to 3 (high reflection), where 1.8% of the essays are rated as level 0, 2.8% as level 1, 27.4% as level 2 and 68% as level 3.

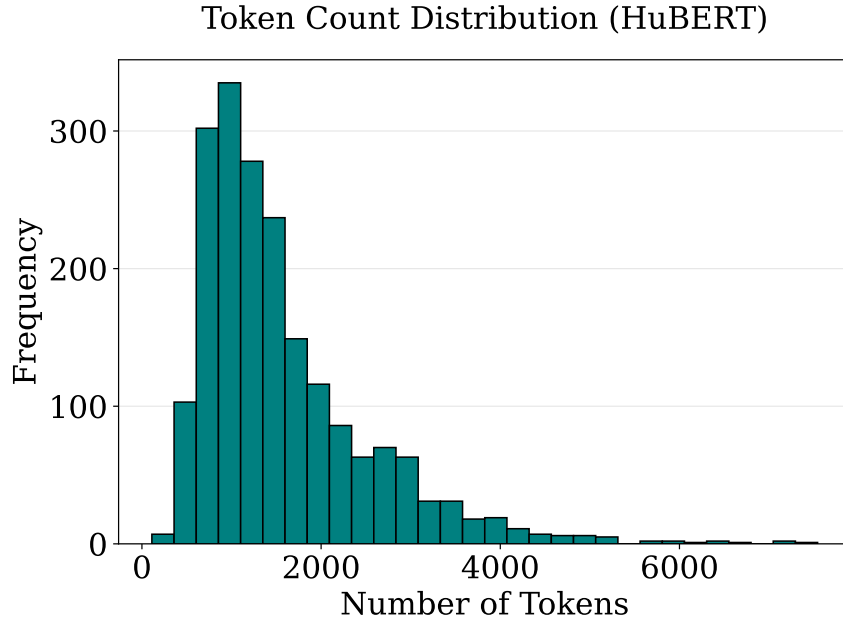


Figure 2: Token count distribution generated by the HuBERT tokenizer. The histogram shows a right-skewed distribution, with most samples containing fewer than 2,000 tokens and peaks around 1,000 tokens.

3.2 Data Preprocessing

The essays from the dataset went through an extensive preprocessing phase to make them suitable for the text processing algorithms (Shallow Machine Learning, Transformer models).

Firstly, the essays came in different formats (docx, pdf, txt) and the text had to be extracted from these documents. The text extraction was done using the `python-docx` (Canny, 2025) Python Library for docx files and `PdfPlumber` (Singer-Vine, 2025) Python library was used for pdf files¹.

Additionally, the documents contained some cover pages as template which are not relevant for the task, so these were removed from the text manually. Also, personally identifiable information was present in the essays, such as names, places, organizations, dates, etc. These were removed with a three-fold approach: first using `huspacy` (SzegedAI, MILAB, 2024) python library with their best model possible, namely `hu_core_news_trf_xl`

¹PyPDF2 (Fenniak, 2022) was also tried, but it had issues with Hungarian characters and inserted random spaces inside words.

a model built on XLM-RoBERTa-large architecture for named entity recognition removing most of the names, organisations and places and replace them with a generic placeholder like [PER], [LOC], [ORG]. In the second step, the text was cleaned up by removing whitespaces, new lines, tabs, unnecessary punctuation marks and spaces. Finally, the text was cleaned using a custom regular expression (for more details see Additional File 3) to remove dates, month names, roman numerals, and specific Hungarian words which could contain locality information (for example word "kerület" which means district in Hungarian and it was not covered by the named entity recognition).

As a last step, the texts were manually checked again for any remaining sensitive information and saved into a dataset format called `.arrow` (Apache Arrow, 2023) which is a columnar format that is suitable for large datasets and efficient for processing with machine learning libraries like HuggingFace’s `datasets`.

3.3 Shallow Machine Learning Models

The most commonly used models in the literature for reflection detection are the following: Random Forest (RF), XGBoost, CatBoost and Ridge-Classifier. These models were tested on the Hungarian essay dataset used in this study. The features for these models in every case were selected from different sources:

- TFIDF features: using the `TFIDFVectorizer` from the `sklearn` Python library, which calculates the term frequency-inverse document frequency of the words in the essays, makes a constant length vector representation of the essays, which can be used by the models.
- Document embeddings: using pre-trained document embeddings to represent the essays. Based on the Massive Text Embeddings benchmark one of the best model in the multi-lingual category is the `Qwen3-4B-Embedding` by Zhang et al. (2025) model. Qwen3 embeddings gives a fixed-length 2560-dimensional vector representation of the essays, which can be used by the models.

The models were trained using the `sklearn` library and the training was done with a 5-fold cross-validation in each case using `StratifiedKFold`. The random seed for all experiments was set to 42.

The data was split into 80% training and 20% testing split on the essay level, and the results were evaluated using the following metrics: **Accuracy**,

Precision, Recall, F1-score, ROC-AUC. The results were visualized using Confusion Matrices and ROC Curves.

As outlined in Figure 1, the Hungarian essays dataset is heavily imbalanced. Firstly the models were trained on the original data without any balancing techniques to obtain the baseline results.

After that multiple balancing techniques were evaluated:

- Class weighting: using the `class_weight` parameter in the models to give more weight to the minority classes and less weight to the majority classes. XGBoost uses sample weights instead of the class weights.
- Oversampling techniques: using multiple oversampling techniques SMOTE (Chawla et al., 2002) (Synthetic Minority Over-sampling Technique) and ADASYN (He et al., 2008) (Adaptive Synthetic Sampling), RandomOverSampling (Japkowicz and Stephen, 2002) and, BorderlineSMOTE (Han et al., 2005) from the `smote-variants` (Kovács, 2024) Python library to generate synthetic samples from the minority classes.
- Ensemble techniques with oversamplers: using `EasyEnsembleClassifier` (Liu et al., 2008), `SMOTEBoostClassifier` (Chawla et al., 2003), `SMOTEBaggingClassifier` (Wang and Yao, 2009), `AdaUBoostClassifier` (Sun et al., 2007), and `CompatibleAdaBoostClassifier` (Beja-Battais, 2023) from the `imbalanced-ensemble` (Liu, 2025) Python library, to combine the oversampling techniques with ensemble methods.

All the results coming from these models were K-fold cross-validated with 5 folds, and the results were averaged over the folds.

3.4 Transformer Models

Transformer models are powerful deep learning models that have revolutionized the field of natural language processing (NLP) in recent years. The training of transformer models in Hungarian language is challenging due to the limited availability of large-scale annotated datasets and pre-trained models specifically designed for Hungarian. However, several transformer models have been developed that support Hungarian language. From these models, the following were used for this paper:

- `SZTAKI-HLT/hubert-base-cc`: Nemeskey (2021) introduced a BERT-based model trained on a Hungarian subset of the Common Crawl and a snapshot of the Hungarian Wikipedia.

- **NYTK/PULI-BERT-Large:** The work of Yang et al. (2023), they introduced a Hungarian BERT large model based on MegatronBERT.

Both hubert and PULI-BERT-Large have a short context window (512 tokens), which is a limitation for document level classification tasks. To address this limitation, chunking with overlap was applied (128 token for chunking and 64 token for overlap which gives a good performance-efficiency as shown by Jaiswal and Milios (2023)) to split the essays into smaller chunks and then aggregating the chunk level predictions using average pooling.

These models were fine-tuned on the Hungarian essay dataset for the task of reflection level classification. For each model, the pre-trained base model was loaded from the HuggingFace model hub using the **transformers** Python library. Then a classification head was added on top of the base model to perform document level reflection classification on the 4 categories. The base model weights were frozen in some cases to reduce the number of trainable parameters or not frozen to allow the model to learn the specific task better.

To handle data imbalance during the training of the transformer models, multiple techniques were evaluated:

- **Oversampling:** Random oversampling was used to make more samples for the minority classes. To make the new samples by oversampling, the RandomOverSampler (ROS) from the **imblearn** Python library was used.
- **Text Augmentation:** Using text augmentation techniques like back-translation to generate similar samples for the minority classes. During backtranslation only those samples were considered which had at least 80% embedding similarity with the original text to avoid generating too different samples. To make the backtranslated data, English was used as the pivot language. For the translation to English and back to Hungarian the **Helsinki-NLP/opus-mt-hu-en** (Tiedemann et al., 2024) and **Helsinki-NLP/opus-mt-en-hu** (Tiedemann et al., 2024) models from HuggingFace were used. For calculating the embedding similarity the **sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2** (Reimers and Gurevych, 2019) model was used.
- **Loss Functions:** Besides the standard Cross Entropy Loss, Weighted Cross Entropy Loss, Focal Loss and Dice loss were also tried to give more importance to the minority classes during training.

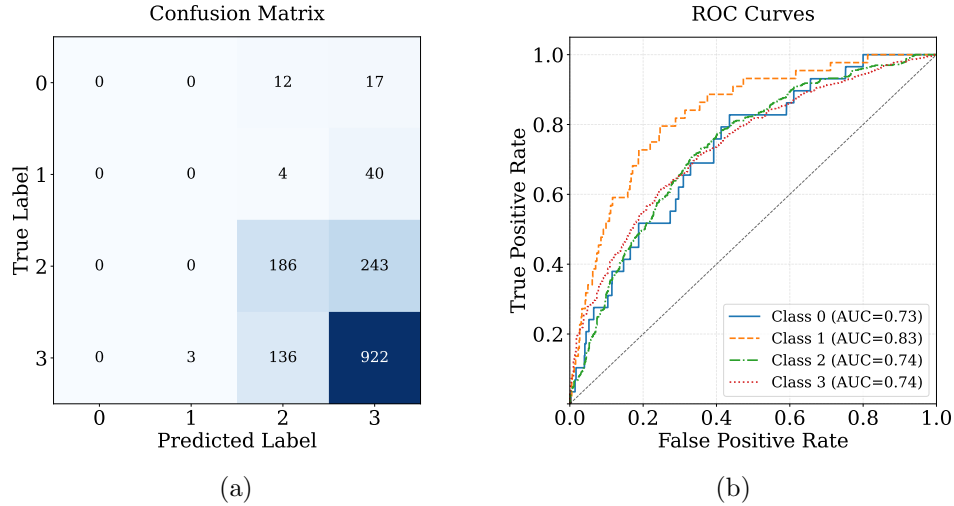


Figure 3: Confusion matrix (a) and ROC curves (b) of the best Shallow Machine Learning model (SMOTEBoost + Qwen3 embeddings + Decision tree with 12 maximum depth) on the Hungarian essays dataset. The results are averaged over 5 folds.

The training pipeline was implemented using the **transformers** (Hugging Face team, 2025) and **datasets** (HuggingFace Inc., 2025) Python libraries from HuggingFace. Early stopping was used based on the **validation accuracy** and patience of 5 epochs allowing the model to stop training early and prevent overfitting. During training linear scheduler, AdamW optimizer were used with weight decay and the random seed was 42. The experiments were run on a single NVIDIA 3090 GPU with 24GB of VRAM.

As the shallow machine learning models, the transformer models were also trained using an 80% training and 20% testing split, and the results were calculated with stratified 5-fold cross/validation using the same metrics as described in 3.3.

4 Experimental Results

In this section the results of the experiments are presented. First the the results of the Shallow Machine Learning models are described then finally the results of the Transformer models.

Model	OS.	Qwen3-E.	Accruacy	F1	ROC Auc	Overall
EasyEnsemble	✓	✓	0.6320	0.6481	0.7240	0.6680
AdaUBoost	✓	✓	0.6967	0.6197	0.7426	0.6863
XG-Boost	✗	✓	0.7082	0.6668	0.7178	0.6976
CompatibleAdaBoost	✓	✓	0.7075	0.6568	0.7364	0.7003
Random Forest	✗	✓	0.7056	0.6613	0.7360	0.7010
CatBoost	✗	✓	0.7012	0.6540	0.7480	0.7010
SMOTEBagging	✓	✓	0.6973	0.6623	0.7569	0.7055
RidgeClassifier	✗	✓	0.7223	0.6833	0.7339	0.7131
SMOTEBoost	✓	✓	0.7088	0.6808	0.7633	0.7176

Table 1: Best result of each model using the Hungarian essay dataset. The results are averaged over 5 folds. The best results in each column are in bold. Green checkmark (✓) indicates the usage of the specific technique, while red cross (✗) indicates its absence. The abbreviation OS. stands for Oversampling, while Qwen3-E. stands for Qwen3 Embeddings.

4.1 Shallow Machine Learning models

As presented in 3.3, multiple Shallow Machine Learning models with multiple different imbalanced data handling techniques were trained and evaluated on the training dataset. The results of the models are presented in Table 1. This is not the complete table (for full table see Additional File 1 from the supplementary), it only indicates the best result of each model with the best hyperparameters (like oversampling technique, usage of semantic embeddings etc.).

As summarized in Table 1, the best results were achieved by the Smote-BoostClassifier which are using the SMOTE oversampling model, as a base estimator it uses a Decision tree with 12 maximum depth and taking the Qwen3 semantic embeddings. This model achieved an average score of **0.7176** (averaged over the 3 metrics), with 0.7089 mean accuracy (with 0.0263 standard deviation), 0.6808 mean F1-score (with 0.0309 standard deviation) and 0.7633 mean ROC-AUC score (with 0.0285 standard deviation) averaged over the 5 folds.

The confusion matrix of the best model (SMOTEBoost + Qwen3 embeddings + Decision tree with 12 maximum depth) is shown in Fig. 3a. The confusion matrix indicates higher performance on majority classes, while minority-class samples are frequently misclassified as majority-class samples. Even misclassified 3 samples from the majority class 3 as minority class 1.

The ROC curve for each class with the best Shallow Machine Learning

Method name	Accuracy		F1		ROC AUC		Overall	
	✓	✗	✓	✗	✓	✗	✓	✗
Class balancing	0.6408	0.6672	0.6283	0.6404	0.7083	0.7134	0.6591	0.6736
Oversampling	0.6481	0.6724	0.6358	0.6294	0.7087	0.7176	0.6604	0.6731
Semantic Embeddings	0.5977	0.6259	0.5891	0.5772	0.6627	0.6446	0.6165	0.6159

Table 2: Ablation study results of different techniques on the Hungarian essays dataset. The results are averaged over the models which were 5-fold cross-validated. Green checkmark (✓) indicates the usage of the specific technique, while red cross (✗) indicates its absence. The best results in each row are in bold.

model is shown in Fig. 3b. The ROC curves show that the SMOTEBoost model can distinguish in a balanced way between the classes, as the AUC scores of each class are above 0.73 with a better performance on the class 1.

In general, classical machine learning models which are able to handle imbalanced data well, like XGBoost, RidgeClassifier, Random Forest achieved good results on this dataset. However these models mostly learnt to classify the majority class well, not even predicting the minority classes correctly as shown on the Fig. 3a. The introduction of oversampling/undersampling techniques and class balancing made the models classify the minority classes occasionally, but not necessarily improve the overall performance of the models. Besides the EasyEnsemble model, the introduction of ensemble techniques with oversampling/undersampling, made similar or even better results than the base shallow machine learning models.

Additionally, in some cases handling the imbalance made the models worse because they started to misclassify the majority classes heavily. One example when this happened is with the RidgeClassifier. It produced a good ROC-AUC score, but does not necessarily mean high performance in the other metrics as well like accuracy and F1-score. For example RidgeClassifier using Qwen3 embeddings with alpha parameter of 5, no class balancing and SMOTE oversampling technique achieved 0.7131 ROC-AUC score (with 0.0379 standard deviation), but only 0.5162 accuracy (with 0.0356 standard deviation) over the 5 fold. The ROC-AUC curves and confusion matrix of this model are shown in Fig. 4a and Fig. 4b.

In Table 2 the results of an ablation study are presented. This shows the average performance of the models when using or not using a specific technique like class balancing, oversampling and semantic embeddings. One finding is that the models without class balancing achieved better results in every metric compared to the models with class balancing. In this dataset,

models without class balancing obtained higher average scores across these metrics, as the models are able to learn the class distribution well without any additional techniques and class balancing made the models misclassify more samples from the majority class.

Similarly, oversampling techniques also did not necessarily improve the overall performance of the models. Oversampling is associated with higher F1, while non-oversampled settings show higher Accuracy and ROC-AUC. The usage of semantic embeddings improved the ROC-AUC scores of the models significantly, however the accuracy and F1-scores were better without using semantic embeddings. This shows that semantic embeddings can help the models to distinguish between the classes better. These techniques can help the models to classify the minority classes better, however they do not necessarily improve the overall performance of the models.

Overall, the ablation study shows that the effectiveness of these techniques can vary depending on the specific dataset and model used. It is important to note that due to the high class imbalance in the dataset, the models without any balancing techniques are mostly biased towards the majority class, which can lead to higher accuracy but lower performance on minority classes. The usage of these techniques made the models to at least classify some samples from the minority classes correctly, which is important in imbalanced classification tasks.

4.2 Transformer models

As mentioned in Section 3.4, two Hungarian pre-trained transformer models were supervised fine-tuned on the training dataset. The best results of the selected models and selected hyperparameters (oversampling/data augmentation, different loss functions and backbone freezing) are presented in Table 3 (for more results see Additional File 2).

Generally the PULI-BERT-Large model achieved better results than the hubert-base-cc in the same hyperparameter settings. The best overall results were achieved by the PULI-BERT-Large model using Cross-Entropy loss function with backtranslation technique, achieving an average score of **0.6872** (averaged over the 3 metrics), with 0.7095 mean accuracy (with 0.0123 standard deviation), 0.6762 mean F1-score (with 0.0244 standard deviation) and 0.6760 mean ROC-AUC score (with 0.0196 standard deviation).

Despite PULI-BERT-Large being better in overall score, hubert shows higher minority-class sensitivity in the reported confusion matrices and ROC curves. A clear difference between the two models can be observed in the

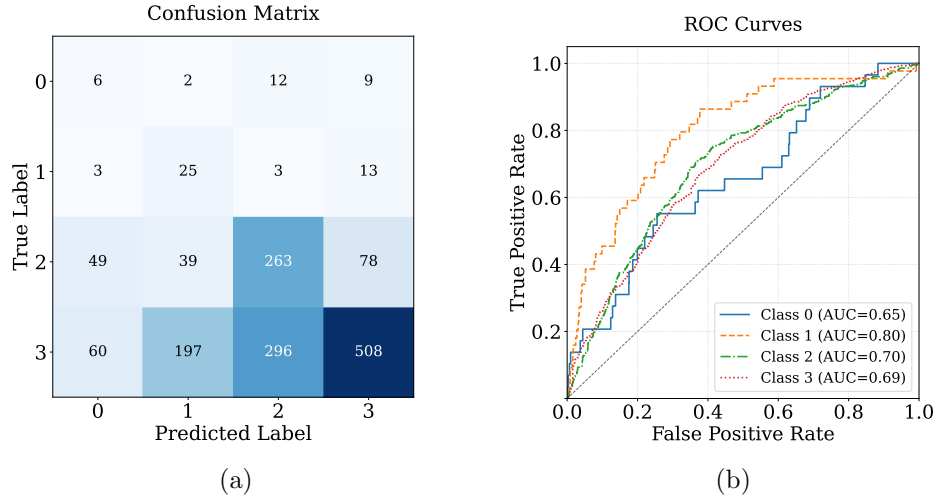


Figure 4: Confusion matrix (a) and ROC curves (b) of an example where RidgeClassifier produces high ROC-AUC score but low accuracy on the hungarian essays dataset. The results are averaged over 5 folds.

Model	hubert-base-cc	PULI-BERT-Large
Loss function	Focal	Cross-Entropy
Oversampling/Backtranslation	✓	✓
Backbone freeze	✗	✗
Accuracy	0.6967	0.7095
F1	0.6601	0.6762
ROC Auc	0.6962	0.6760
Overall	0.6843	0.6872

Table 3: Best result of each Transformer model using the Hungarian essay dataset. The results are averaged over 5 folds. The best results in each column are in bold. Green checkmark (✓) indicates the usage of the specific technique, while red cross (✗) indicates its absence.

ROC curves and confusion matrices in Fig. 5. The ROC curves show worse performance from both transformer models compared to the best Shallow Machine Learning models, however the transformer models are able to classify the minority classes better, as shown in the confusion matrices in Fig. 5c and Fig. 5d.

The experiments with different loss functions are presented in Table 4. The best results were achieved by the Cross-Entropy Loss, which is the

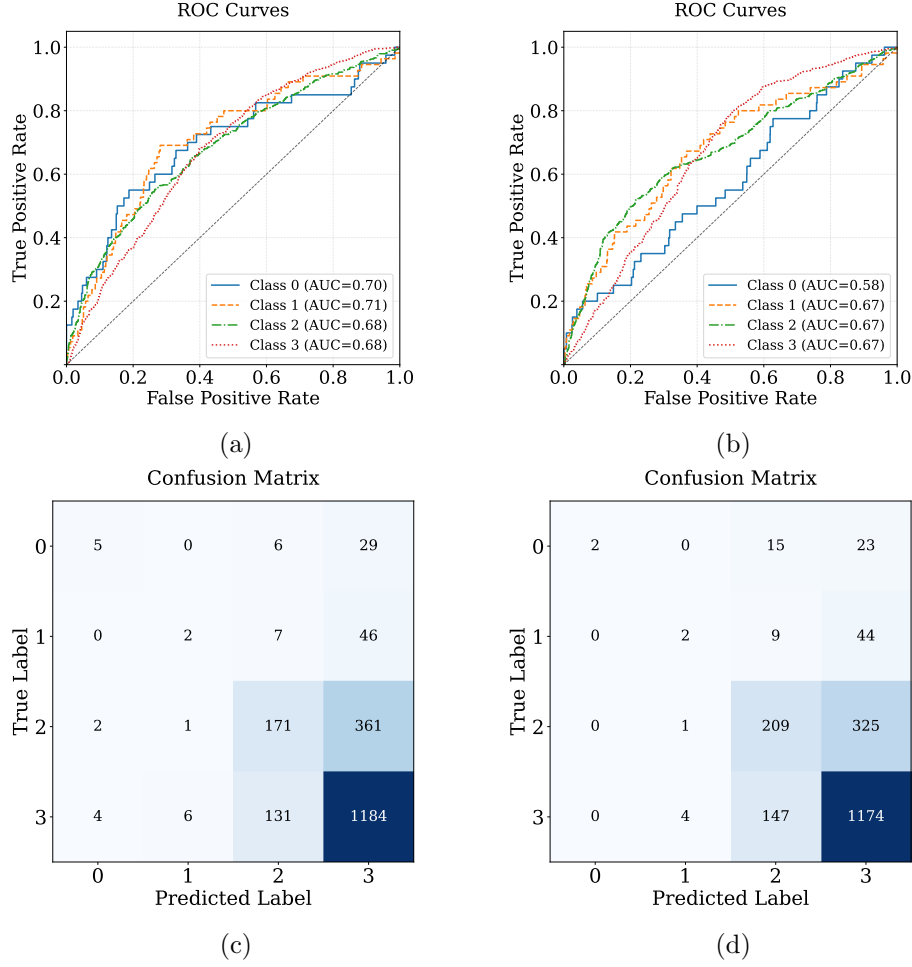


Figure 5: (a) ROC Curve of the Hubert model with backtranslation and focal loss and (b) ROC Curve of the PULI-BERT-Large model with backtranslation and cross-entropy loss. (c) Confusion Matrix of the Hubert model with backtranslation and focal loss and (d) Confusion Matrix of the PULI-BERT-Large model with oversampling and cross-entropy loss. The results are averaged over 5 folds.

standard loss function for classification tasks, yields the highest aggregate performance among the tested losses. The Dice Loss achieved similar results, while the Focal loss and Weighted Cross-Entropy Loss achieved worse results.

The results of an ablation study on the transformer models are presented

Losses	Accuracy	F1	ROC-AUC	Overall
Cross-Entropy Loss	0.5614	0.5361	0.6217	0.5731
Dice Loss	0.5670	0.5529	0.5951	0.5716
Focal Loss	0.4800	0.4523	0.6029	0.5118
Weighted Cross-Entropy Loss	0.4822	0.4502	0.6044	0.5123

Table 4: Results with different loss functions, metrics were averaged over 2 transformer models and different hyperparameters (like oversampling and data augmentation). The best results in each column are in bold.

Method	Accuracy	F1	ROC-AUC	Overall
ROS	0.5784 (27% ↑)	0.5914 (30% ↑)	0.6389 (8% ↑)	0.6029 (22% ↑)
Backtranslation	0.6859 (38% ↑)	0.6119 (32% ↑)	0.6259 (7% ↑)	0.6412 (26% ↑)
Freeze Encoder	0.4476 (-15% ↓)	0.4247 (-14% ↓)	0.5634 (-8% ↓)	0.4786 (-12% ↓)

Table 5: Ablation study results of different techniques on the transformer models using the Hungarian essays dataset. The results are averaged over the 2 transformer models and different hyperparameters. The values in parentheses indicate the relative change compared to the baseline model without using the specific technique. Green values with upward arrows indicate improvement, while red values with downward arrows indicate performance decrease.

in Table 5. The table shows the average performance of the models when using or not using a specific technique like oversampling, data augmentation and freezing the backbone during training. As the table shows, both oversampling and data augmentation are associated with higher average transformer performance in this study. Compared to shallow models, oversampling and data augmentation have a more significant positive impact on the performance of the transformer models. However, freezing the backbone during training significantly decreased the performance of the models, which shows the importance of full fine-tuning when adapting transformer models to specific tasks.

The overall results of the transformer models are little lower than the best Shallow Machine Learning models. The best results with the transformer models are around $\approx 68\%$ overall score averaged over accuracy, F1-score, and ROC AUC metrics. However the transformer models are able to classify the minority classes better, as the confusion matrix in Fig. 5c and Fig. 5d presents. Confusion matrices indicate improved minority-class detection for transformer models compared to shallow baselines in these settings.

However, due to the high class imbalance, most minority class samples are still misclassified as the majority classes.

4.3 Statistical significance test

In this section the results of the statistical significance test are presented. The T-test was used to compare the performance of the best shallow machine learning model (SMOTEBoost + Qwen3 embeddings + Decision tree with 12 maximum depth) and the best transformer model (PULI-BERT-Large with backtranslation and cross-entropy loss) on the Hungarian essays dataset. The T-test was performed on the Accuracy, F1-score, ROC-AUC score and Macro-F1 score of the two models across the 5 folds.

The results show that there are no statistically significant differences between the two models in terms of Accuracy ($t = -0.0464$, $p = 0.9652$) and Weighted F1-score ($t = 0.2862$, $p = 0.7889$).

Similarly, no statistically significant difference was observed for the Macro-F1 score ($t = -0.9815$, $p = 0.3819$).

In contrast, a statistically significant difference was found for ROC-AUC ($t = 5.5053$, $p = 0.0053$), where the shallow model achieved higher scores than the transformer model.

5 Discussion

The findings indicate that automatic reflective text classification in Hungarian achieves performance comparable to previously reported results in other languages (Zhang et al., 2024; Nehyba and Štefánik, 2023). Shallow models obtained higher aggregate means, while statistical tests indicated no significant differences for Accuracy and Weighted F1. The difference remains modest, suggesting that both paradigms are capable of capturing key characteristics of reflective writing. While shallow models can distinguish between classes better, transformer models show improved sensitivity to minority classes, which is relevant for imbalanced educational datasets. This study addresses document-level reflective text classification for Hungarian essays, highlighting the feasibility of this task in a morphologically rich language.

The findings further emphasize the importance of handling class imbalance. Techniques such as oversampling, class weighting, different loss functions, and data augmentation were particularly beneficial, especially for transformer-based models. This suggests that transformer models are more

sensitive to class distribution and can benefit substantially from strategies that improve the representation of minority classes.

These results may inform educational tools for preliminary feedback and workload support, with final assessment retained by educators. In particular, the ability of models to accurately assess reflective quality in writing can be leveraged in educational technologies, where automated feedback on student essays can enhance learning outcomes and the development of critical thinking skills.

Furthermore, these systems can reduce teachers’ workload by providing initial assessments of student essays, reducing subjectivity and allowing educators to focus on more personalized feedback and instruction. The findings also suggest that integrating both shallow and deep learning approaches could yield hybrid models that use the strengths of each method, potentially leading to even more effective reflective analysis systems.

While our approach reports results for Hungarian reflective text classification, several limitations must be acknowledged. First, the effectiveness of these methods depend on class distribution and annotation quality. Even with oversampling, most models struggled to accurately classify minority classes, often achieving high overall accuracy by favoring majority class predictions. This highlights the need for further research into techniques that can better handle imbalanced datasets, particularly in the context of reflective analysis in Hungarian language essays. Future work could explore more advanced data augmentation strategies or the collection of more balanced datasets.

Another limitation arises from the restricted context window of transformer models, which makes the prediction more difficult due to the information loss when truncating longer essays into smaller chunks and aggregate them. Addressing this issue with Hungarian specific or multi lingual large language models that can handle longer contexts could further improve performance.

6 Conclusion

This study examined automatic reflection-level classification in Hungarian student essays using a large expert-annotated corpus (N=1,954). This study provides a comprehensive document-level comparison in this Hungarian setting between shallow machine-learning methods and Hungarian pre-trained transformer models under severe class imbalance.

The main finding is a consistent trade-off between aggregate perfor-

mance and minority-class behavior. Shallow models with strong feature representations achieved the best overall scores (approximately 71% aggregate score over Accuracy, F1, and ROC-AUC), while transformer models achieved slightly lower overall performance (approximately 68%) but provided more balanced behavior on underrepresented reflection levels. This indicates that, in low-resource and imbalanced educational NLP settings, classical approaches remain highly competitive, while transformer models are valuable when minority-class sensitivity is a priority.

Our experiments also show that imbalance-handling methods are not universally beneficial: their effect depends on model family and objective metric. In several configurations, these techniques improved minority-class detection but reduced majority-class stability, reinforcing the need for metric-aware model selection.

The study has two main limitations. First, the strong class skew remains a bottleneck for robust minority-class learning. Second, transformer context limits required chunking long essays, which can weaken document-level coherence signals. Future work should therefore focus on better minority-class data support (targeted collection and augmentation), longer-context Hungarian or multilingual architectures, and hybrid systems that combine the strong global discrimination of shallow models with the minority-class robustness of transformer-based methods.

List of abbreviations

LLM - Large Language Model
OECD - Organisation for Economic Co-operation and Development
RWF - Reflective Writing Framework
PoS - Part of Speech
STEM - Science, Technology, Engineering and Mathematics
CS - Computer Science
NLP - Natural Language Processing
TF-IDF - Term Frequency-Inverse Document Frequency
RF - Random Forest
ROC-AUC - Receiver Operating Characteristic - Area Under Curve
Qwen3-E - Qwen3 Embeddings
OS - Oversampling
SMOTE - Synthetic Minority Over-sampling Technique
ROS - Random Oversampling

Declarations

Data availability

Due to ethical considerations, the datasets used and/or analyzed during the current study cannot be made publicly available and are not accessible.

Competing interests

The authors declare that there are no conflicts of interest related to the research, authorship, and publication of this article.

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